Ch 5: Life Processes

SECTION A - LIFE PROCESSES, CHARACTERISTICS OF LIVING THINGS AND INTRODUCTION TO NUTRITION

1 Meaning of Life Processes

Life processes are the **vital activities** carried out by living organisms to stay alive, grow, reproduce and respond to their **surroundings**.

2 Why Maintenance Processes Are Necessary

- Living organisms constantly use **energy**.
- Body parts wear out and need **repair**.
- Waste products are formed and must be **removed**.
- To maintain **order**, **balance** and proper functioning of the body.
- These processes keep the organism **alive** and **active**.

3 Living vs Non-living Things

Living Things	Non-living Things
• Have internal molecular movements. • Require food, water, oxygen. • Carry out maintenance processes. • Grow, reproduce and respond to stimuli.	• No internal molecular movements. • Do not need food, water or oxygen. • No maintenance processes. • Do not grow or reproduce.

4 Main Life Processes

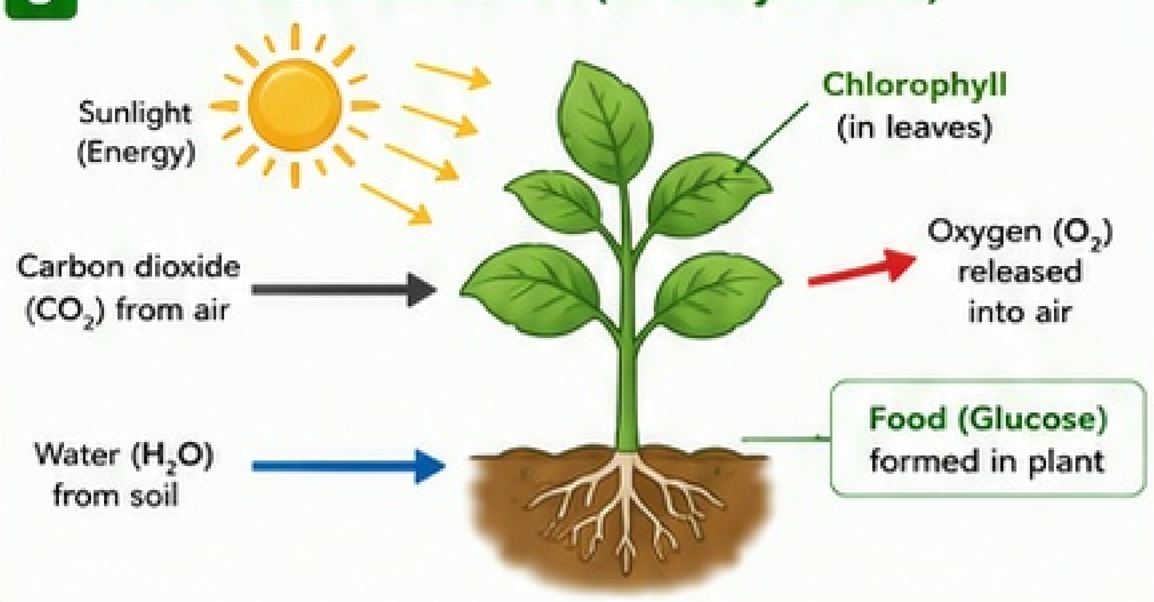
The four main life processes in all living organisms are:

Nutrition | Respiration | Transportation | Excretion

5 Role of Each Life Process (Brief)

- **Nutrition** – Provides energy anol materials for growth, repair and maintenance.
- **Respiration** – Releases energy from food for all activities.
- **Transportation** – Carries nutrients, gases and wastes to and from different parts of the body.
- **Excretion** – Removes harmful waste products from the body.

9 How Plants Make Food (Photosynthesis)



6 Meaning of Nutrition

Nutrition is the process by which organisms take in food and use it for energy, growth, repair and maintenance of the body.

7 Types of Nutrition

- **Autotrophic Nutrition** – Organisms make their own food from simple inorganic substances.
  Examples: Green plants, algae, some bacteria.
- **Heterotrophic Nutrition** – Organisms cannot make their own food and depend on other organisms for food.
  Examples: Animals, fungi, most bacteria.

8 Overview: Food Dependence in Living Organisms

PLANTS
(Autotrophs)
Prepare their own food.

↓

ANIMALS (Heterotrophs)
Cannot prepare their own food.

↓ ↓

DIRECTLY DEPEND ON PLANTS
Herbivores eat plants.
Example: Cow, Deer, Rabbit

INDIRECTLY DEPEND ON PLANTS
Carnivores eat other animals which eat plants.
Example: Lion, Eagle

Thus, all animals depend directly or indirectly on plants for food.

10 Important Definitions

Autotroph: Organism that can produce its own food from simple inorganic substances using light or chemical energy.

Heterotroph: Organism that cannot make its own food and depends on other organisms for food.

Life Process: Any vital activity necessary for the survival and maintenance of a living organism.

Nutrition: The process of taking in food and using it for energy, growth, repair and maintenance.

11 Autotrophic vs Heterotrophic Nutrition

Basis	Autotrophic Nutrition	Heterotrophic Nutrition
Ability to make food	Can make their own food.	Cannot make their own food.
Raw materials used	CO_2, H_2O and minerals.	Organic food from other organisms.
Energy source	Sunlight (in plants) or chemical energy (in some bacteria).	Energy stored in organic food obtained.
Examples	Green plants, algae, some bacteria.	Animals, fungi, most bacteria.
Role in ecosystem	Producers	Consumers

QUICK RECALL

- Life processes keep us alive.
- Four main life processes: Nutrition, Respiration, Transportation, Excretion.
- Plants make food; animals depend on plants.

KEY IDEA

All life processes work together to maintain life. Nutrition is the first and most important step, as it provides the energy and materials needed for all other activities.

EXAM TIPS

- Learn the four life processes and their roles clearly.
- Differentiate autotrophic and heterotrophic nutrition with examples.
- Draw and label the photosynthesis diagram neatly.

CENTUM FOCUS

Create a flow chart of food dependence (plants → animals) and write key differences in a table. Practice diagram and definition writing for better exam scoring.

CLASS 10 SCIENCE

Ch 5: Life Processes

Aim High. Score Centum!

SECTION B - AUTOTROPHIC NUTRITION IN PLANTS AND PHOTOSYNTHESIS

1 Autotrophic Nutrition – Definition

- Mode of nutrition in which organisms (green plants) prepare their own food from simple inorganic substances using sunlight in the presence of **chlorophyll**.

2 Photosynthesis – Meaning

- The process by which green plants prepare food (glucose) from carbon dioxide and water using sunlight in the presence of chlorophyll is called **photosynthesis**.

3 Equation for Photosynthesis

(i) Word Equation	Carbon dioxide + Water $\xrightarrow[\text{Chlorophyll}]{\text{Sunlight}}$ Glucose + Oxygen
(ii) Balanced Symbolic Equation	$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$

4 Raw Materials Needed

- **Carbon Dioxide (CO_2)** – Enters through stomata.
- **Water (H_2O)** – Absorbed by roots and transported to leaves.
- **Sunlight** – Trapped by chlorophyll for energy.
- **Chlorophyll** – Green pigment that traps light energy.

5 Where is Chlorophyll Present?

- Chlorophyll is present in **chloroplasts**, which are found in the green parts of the plant, especially in the **mesophyll cells** of leaves.

6 Stomata and Their Role

- Stomata are tiny pores present mainly on the lower surface of leaves.
- **Guard cells** control the opening and closing of stomata.
- Role: Allow CO_2 to enter the leaf for photosynthesis and permit O_2 and water vapour to exit.

7 How CO_2 Enters and Water Reaches Leaves

- CO_2 enters leaves through stomata by diffusion.
- Water absorbed by roots from soil is transported through **xylem** vessels to leaves.

8 Role of Roots and Xylem

- Roots absorb water and minerals from soil.
- Xylem vessels transport water and minerals upward to leaves.

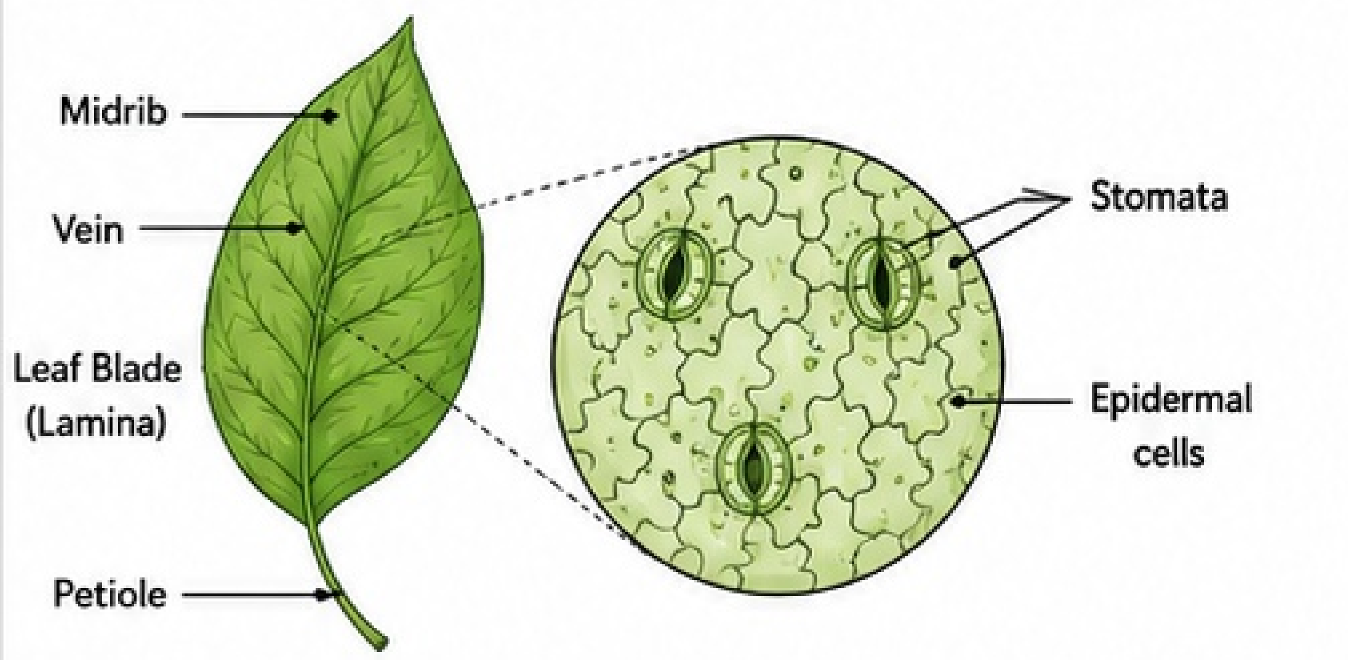
9 By-product of Photosynthesis

- **Oxygen (O_2)** is released as a by-product through stomata.

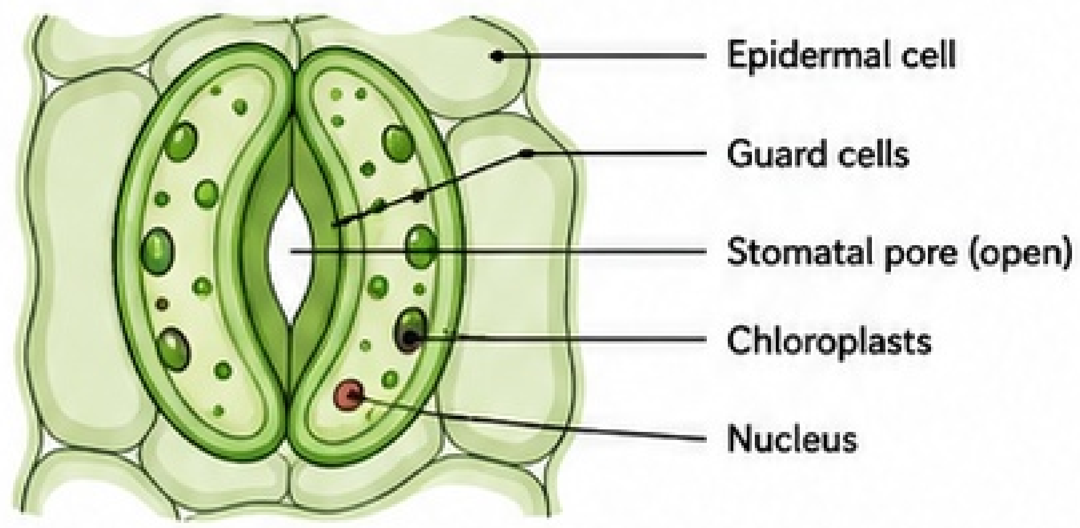
10 Factors Affecting Photosynthesis

Factor	Effect
Light	Increase in light intensity increases rate up to a limit.
Chlorophyll	More chlorophyll → more light absorption → higher rate.
CO_2 Concentration	Increase in CO_2 increases rate up to a limit.
Water	Water deficiency slows down photosynthesis.

11 Leaf – Showing Stomata (Lower Epidermis View)



12 Stomatal Apparatus (Opened)



13 Destarching and Starch Test (Idea Only)

- **Destarching:** Plant is kept in dark for 24–48 hours to use up stored starch.
- **Starch Test:** Leaf is boiled in water, then in alcohol, rinsed and treated with iodine solution.
- **Observation:** **Blue-black** colour indicates presence of starch → photosynthesis occurred.

14 Heterotrophic Modes in Plants

Mode	Description	Examples
Parasitic	Plants obtain food from a living host and harm it.	Cuscuta (Amarbel), Viscum (Mistletoe)
Saprophytic	Plants obtain food from dead and decaying organic matter.	Bread mould (Rhizopus), Mushroom (Agaricus)
Insectivorous	Plants trap and digest insects to get nutrients, especially nitrogen.	Pitcher plant (Nepenthes), Venus flytrap (Dionaea)
Symbiotic	Two organisms live together and both benefit.	Lichen (Algae + Fungus), Root nodules of legume plants with Rhizobium

KEY TAKEAWAY

- Green plants prepare their own food by photosynthesis.
- $CO_2 + H_2O$ + Light + Chlorophyll → Glucose + O_2
- Stomata, chlorophyll, xylem, roots – all work together.

EXAM TIP

- Always write balanced equation with conditions (sunlight, chlorophyll).
- Diagrams of stomata and leaf fetch important marks.

QUICK REVISE

- Raw materials: CO_2, H_2O, Sunlight, Chlorophyll
- **By-product:** Oxygen
- **Factors:** Light, Chlorophyll, CO_2, Water

CENTUM FOCUS

- Label diagrams neatly.
- Write points in bullet form.
- Use keywords in blue/green.
- Practice SAQ and diagram-based questions.

SECTION C - HETEROTROPHIC NUTRITION AND HUMAN DIGESTIVE SYSTEM

1 Meaning of Heterotrophic Nutrition

- Heterotrophic nutrition is the mode of nutrition in which organisms cannot make their own food and depend on other organisms for ready-made organic food.

2 Types of Heterotrophic Nutrition

(i) Holozoic Nutrition

- Food is taken in solid form, digested inside the body and absorbed.
- Common in animals.
- **Examples:** Amoeba, Humans, Cow, Dog.

(ii) Saprophytic Nutrition

- Organisms feed on dead and decaying organic matter by secreting enzymes externally and absorbing the soluble food.
- **Examples:** Fungi (Rhizopus, Mushroom), Some bacteria.

(iii) Parasitic Nutrition

- Organisms live on or in a host and obtain food at the expense of the host.
- **Examples:** Tapeworm, Leech, Cuscuta (Amarbel), Head louse.

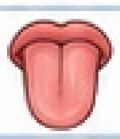
3 Stages of Nutrition in Heterotrophic Organisms

1. **Ingestion:** Taking food into the body.
2. **Digestion:** Breakdown of complex food into simple soluble form.
3. **Absorption:** Absorbing the digested food into blood or body fluids.
4. **Assimilation:** Utilization of absorbed food by cells for energy, growth and repair.
5. **Egestion:** Removal of undigested and unabsorbed food from the body.

6 Functions of Buccal Cavity and Associated Organs

Buccal Cavity (Mouth): Receives food, moistens it with saliva, chews and forms it into a bolus for swallowing.

Teeth: Help in cutting, tearing and grinding food.
Types: Incisors (cut), Canines (tear), Premolars and Molars (grind).



Tongue: Mixes food with saliva, helps in taste, forms bolus and pushes it back for swallowing.

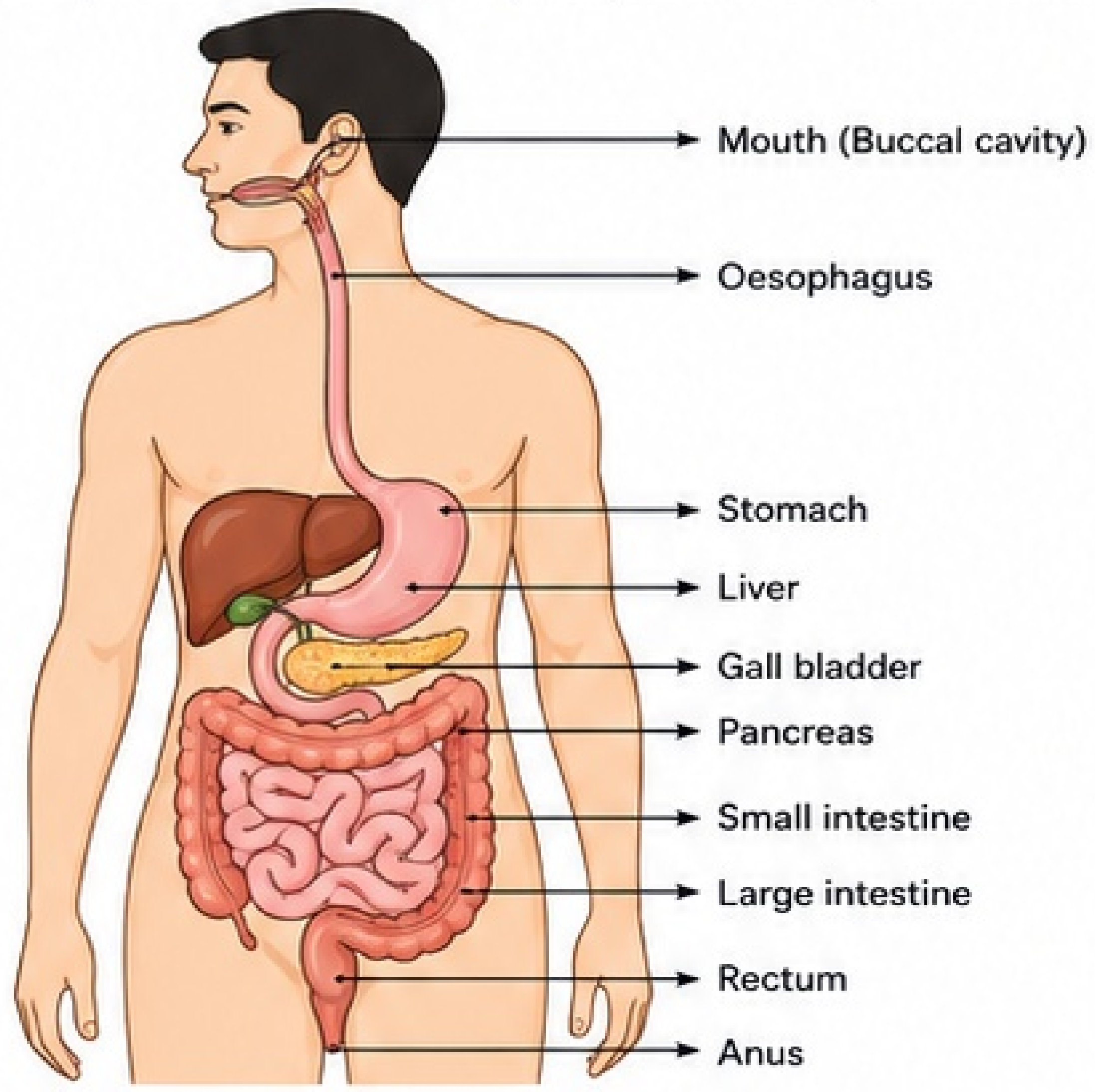
Salivary Glands: Secrete saliva which moistens food and contains enzyme salivary amylase (ptyalin) that starts digestion of starch into simple sugars.

7 Role of Oesophagus and Peristalsis

- **Oesophagus** (food pipe) is a muscular tube connecting mouth to stomach.
- **Peristalsis:** Rhythmic wave-like contractions of muscles in the oesophagus help in pushing the food down into the stomach irrespective of body position.

4 Human Alimentary Canal

- A long muscular tube running from mouth to anus through which food passes. It has different parts, each with a specific function.



- **Mouth (Buccal cavity):** Ingestion, chewing, mixing with saliva.
- **Oesophagus:** Conveys food to stomach by peristalsis.
- **Stomach:** Stores food, churns and digests proteins.
- **Liver:** Produces bile; helps in fat digestion.
- **Gall bladder:** Stores and concentrates bile; releases it into small intestine.
- **Pancreas:** Secretes pancreatic juice (enzymes) into small intestine.
- **Small intestine:** Completes digestion and absorbs nutrients.
- **Large intestine:** Absorbs water and salts; forms faeces.
- **Rectum:** Stores faeces temporarily.
- **Anus:** Egestion of faeces.

8 Structure and Function of Stomach

- J-shaped, muscular bag between oesophagus and small intestine.
- Inner lining has folds and gastric glands.
- Gastric glands secrete:
  - **Hydrochloric acid (HCl):** Provides acidic medium; kills germs; activates enzymes.
  - **Pepsin:** Digests proteins into peptones.
  - **Mucus:** Protects the stomach lining from acid and enzymes.
- Muscular walls churn the food to form semi-liquid chyme.
- Food remains in stomach for 2–4 hours.

9 Flow Chart: From Food Intake to Egestion

Food Intake (Mouth) → Ingestion → Digestion (Mouth, Stomach, Small Intestine) → Absorption (Small Intestine) → Assimilation (By Body Cells) → Egestion (Anus)

10 Mechanical Digestion vs Chemical Digestion

Feature	Mechanical Digestion	Chemical Digestion
Meaning	Breakdown of food into smaller pieces without changing its chemical nature.	Breakdown of complex food into simple substances by enzymes and digestive juices.
How it occurs	By physical actions like chewing, churning, mixing and peristalsis.	By enzymes (e.g., amylase, pepsin, lipase, trypsin) and acids (e.g., HCl).
Where it occurs	Mouth (chewing), Stomach (churning), Intestine (segmentation).	Mouth, Stomach and Small Intestine.
Result	Food is broken into smaller pieces; surface area increases.	Food is converted into simple, soluble substances (e.g., sugars, amino acids, fatty acids, glycerol).
Example	Chewing of roti, Churning in stomach.	Starch → Maltose (by amylase), Protein → Peptones (by pepsin), Fats → Fatty acids + Glycerol (by lipase).

Section D - Human Digestion: Digestive Juices, Absorption and Assimilation

1. What Happens in Stomach?

- Stomach is a J-shaped muscular organ.
- Food from oesophagus enters the stomach.
- Strong muscular walls churn the food and mix it with gastric juice to form a semi-liquid mass called **chyme**.
- Chyme remains in the stomach for 2–4 hours.
- Only partial digestion of proteins takes place here.

2. Roles of Gastric Juice Components

HCl (Hydrochloric acid)	• Creates an acidic medium (pH 1.5–2.0). • Kills harmful microbes. • Activates pepsinogen to pepsin.
Mucus	• Protects inner lining of stomach from HCl and enzymes. • Prevents self-digestion.
Pepsin (from pepsinogen)	• Digests proteins into smaller molecules called peptones and peptides.

3. Liver and Bile

- Liver is the largest gland in the body.
- It produces bile (500–1000 mL/day).
- Bile is stored and concentrated in the gall bladder.
- Bile has no enzymes. It is alkaline.
- **Functions of bile:**
  - Neutralises acidic chyme from stomach.
  - **Emulsifies fats** into tiny droplets (increases surface area for enzyme action).
  - Helps absorption of fats and fat-soluble vitamins (A, D, E, K).
  - Helps in excretion of waste products (bilirubin, cholesterol).

4. Pancreas and Pancreatic Juice

- Pancreas is a gland located below the stomach.
- It secretes **pancreatic juice** (alkaline) into the small intestine.
- Components and their roles:
  - **Amylase** – breaks starch into maltose.
  - **Trypsin** – breaks proteins into peptides.
  - **Lipase** – breaks fats into fatty acids and glycerol.
  - **Nucleasee** – break nucleic acids into nucleotides.
  - **Bicarbonate (HCO_3^-)** – neutralises acidic chyme and provides alkaline medium for enzymes.

5. Digestion in Small Intestine

- Chyme from stomach enters small intestine.
- Receives bile and pancreatic juice.
- Almost complete digestion of carbohydrates, proteins and fats takes place.
- Digestion is completed with the help of intestinal juice.

6. Intestinal Juice

- Secreted by glands in the wall of small intestine.
- Contains the following enzymes:
  - **Maltase** – converts maltose → glucose
  - **Sucrase** – converts sucrose → glucose + fructose
  - **Lactase** – converts lactose → glucose + galactose
  - **Peptidases** – convert peptides → amino acids
  - **Lipase** – converts fats → fatty acids + glycerol
  - **Nucleases** – convert nucleotides → sugars + bases + phosphate

7. Final Products of Digestion

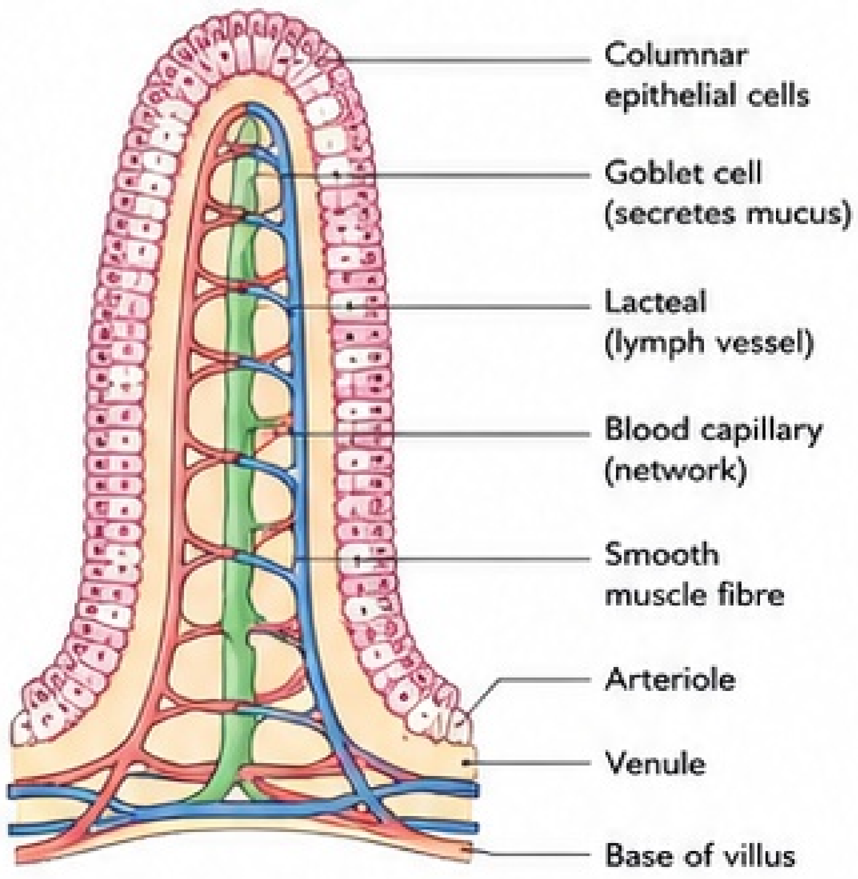
Food	Final Products
Carbohydrates	Glucose (simple sugar)
Proteins	Amino acids
Fats	Fatty acids + Glycerol

11. Why is Small Intestine Long and Coiled?

- It is about 6–7 metres long and highly coiled.
- Reasons:
  - Provides large surface area for maximum digestion and absorption.
  - Food remains in contact with digestive enzymes for a longer time.
  - Ensures efficient absorption of nutrients.

8. Villi and Absorption

- Small intestine is lined with numerous finger-like projections called **villi**.
- Each villus increases surface area for absorption.
- **Structure of a villus:**



How Absorption Occurs:

- Glucose, amino acids, vitamins, minerals and water are absorbed into blood capillaries.
- Fatty acids and glycerol are absorbed into the lacteal (lymph vessel).
- Blood and lymph carry absorbed nutrients to all body cells.

9. Assimilation

- Absorbed nutrients are utilized by body cells for:
  - Energy
  - Growth and repair
  - Maintenance of body functions
  - Storage (extra glucose as glycogen in liver and muscles; fats in adipose tissue)

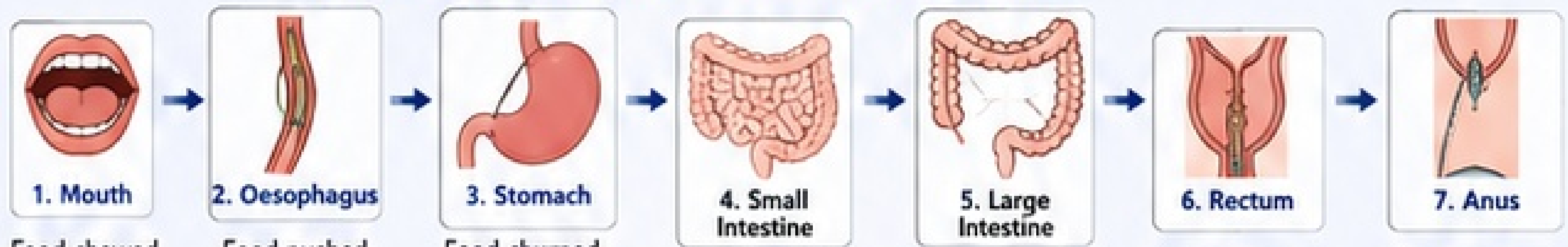
10. Egestion

- The undigested and unabsorbed food (mainly fibre, some water, salts and dead cells) moves to large intestine.
- Water is absorbed here.
- The remaining waste is stored in rectum and removed from the body through anus as faeces.

12. Concise Table: Digestive Organ, Secretion and Main Action

Digestive Organ	Secretion	Main Action
Mouth	Saliva (amylase)	Breaks starch into maltose
Stomach	Gastric juice (HCl, mucus, pepsin)	Partial digestion of proteins; forms chyme
Liver	Bile	Emulsifies fats; neutralises acid; no enzymes
Pancreas	Pancreatic juice (enzymes + bicarbonate)	Digest carbs, proteins, fats, nucleic acids; neutralises acid
Small Intestine (Intestinal glands)	Intestinal juice (enzymes)	Completes digestion; converts all to simple absorptive forms

13. Journey of Food – Stepwise Flow Chart



1. Mouth: Food chewed and mixed with saliva. Starch begins digestion.
2. Oesophagus: Food pushed down by peristalsis.
3. Stomach: Food churned with gastric juice. Partial protein digestion. Forms chyme.
4. Small Intestine: Receives bile and pancreatic juice. Complete digestion with intestinal juice. Nutrients absorbed.
5. Large Intestine: Water absorbed. Waste material forms faeces.
6. Rectum: Faeces stored temporarily.
7. Anus: Faeces removed from body (egestion).

Quick Revision Points

- ★ **Stomach** – protein digestion starts.
- ★ **Bile** – emulsifies fats, no enzymes.
- ★ **Pancreatic juice** – major enzymes + bicarbonate.
- ★ **Small intestine** – complete digestion and absorption.
- ★ **Villi** – increase surface area for absorption.
- ★ Absorbed nutrients → blood/lymph → body cells.

Remember

- ✓ Pepsin works in acidic medium.
- ✓ Bile is alkaline and has no enzymes.
- ✓ Pancreatic enzymes work in alkaline medium.
- ✓ Villi have blood capillaries + lacteal.
- ✓ Carbohydrates → glucose
  Proteins → amino acids
  Fats → fatty acids + glycerol

Exam Focus

- ➢ Roles of HCl, mucus and pepsin.
- ➢ Functions of bile.
- ➢ Components of pancreatic and intestinal juice.
- ➢ Final products of digestion.
- ➢ Diagram of villus – label correctly.
- ➢ Difference between absorption and assimilation.
- ➢ Flow chart of journey of food.

Ch 5: Life Processes

SECTION E – RESPIRATION, BREATHING AND HUMAN RESPIRATORY SYSTEM

1 MEANING OF RESPIRATION

Respiration is a chemical process in which food (glucose) is broken down inside the cells in the presence or absence of oxygen to **release energy**.

2 BREATHING vs RESPIRATION

Feature	Breathing (Physical)	Respiration (Chemical)
Meaning	Taking in oxygen (inhalation) and giving out carbon dioxide (exhalation).	Breakdown of food in cells to release energy.
Nature	Physical process	Chemical process
Where it occurs	In lungs (respiratory system)	In all living cells (cytoplasm/mitochondria)
Energy involved	No energy is produced	Energy is produced and used by the body
Depends on	Respiratory organs	Enzymes in cells
Example	Inhalation, exhalation	Oxidation of glucose

3 TYPES OF RESPIRATION

(A) Aerobic Respiration

Breakdown of food in the presence of oxygen.
Occurs in mitochondria.
Complete oxidation → more energy.

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy (ATP)}$$

(B) Anaerobic Respiration

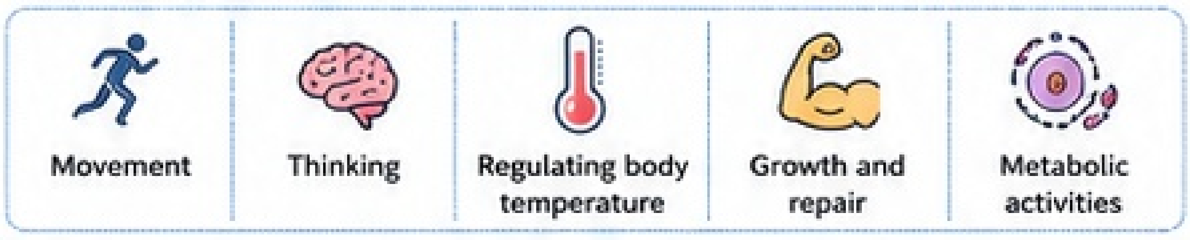
Breakdown of food in the absence of oxygen.
Occurs in cytoplasm.
Incomplete oxidation → less energy.

In yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy}$

In muscles: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + \text{Energy}$ (Lactic acid)

4 RELEASE OF ENERGY FROM FOOD

- During respiration, the bonds in glucose break and energy is released.
- This energy is captured in the form of ATP (Adenosine Triphosphate).
- ATP is the immediate energy currency of the cell.
- The energy is used for:



5 IMPORTANCE OF RESPIRATION

- Supplies energy for all life activities.
- Helps in the breakdown of food into simple substances.
- Maintains body temperature.
- Essential for growth, repair and reproduction.
- Removal of waste gas (CO_2) helps in maintaining internal balance.

6 WHY OXYGEN TRANSPORT IS NECESSARY?

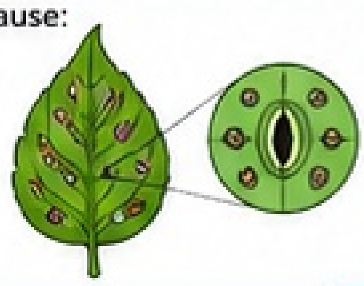
- Oxygen taken in by breathing must reach all cells.
- Blood transports oxygen from lungs to all parts of the body.
- Without oxygen transport, cells cannot do respiration and energy production will stop.

EXAM TIP

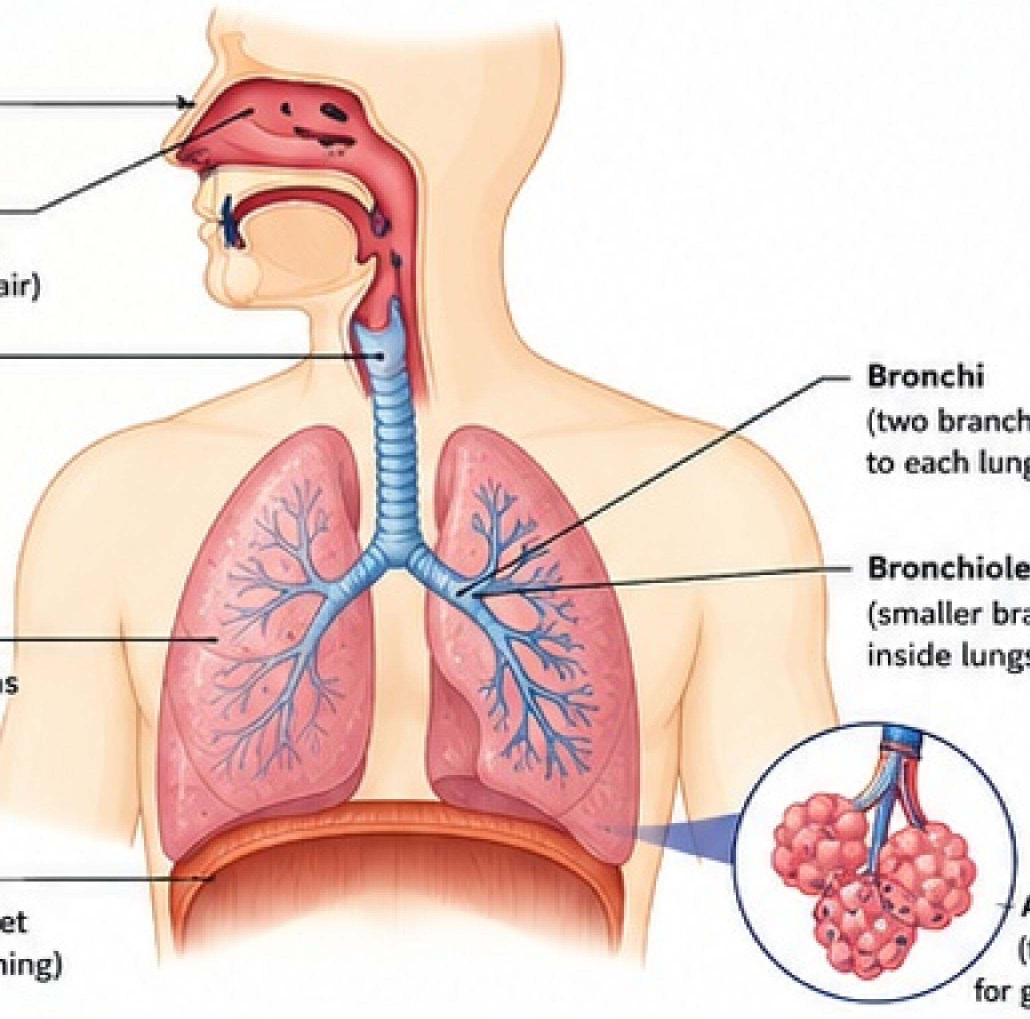
Remember: Breathing = **IN & OUT** (physical)
Respiration = **BREAKDOWN** (chemical in cells)

11 RESPIRATION IN PLANTS (NOTE)

- Plants take in oxygen and release carbon dioxide through stomata in leaves and lenticels in stems.
- Gas exchange in plants is slower than in animals because:
  - Stomata are few and open only at specific times.
  - Plant body is not well ventilated.
  - Diffusion path is longer in plants.



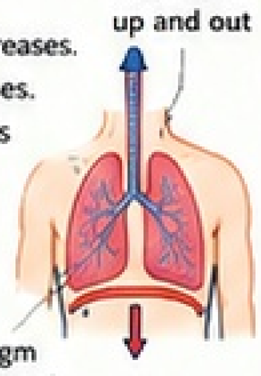
7 HUMAN RESPIRATORY SYSTEM (LABELLED DIAGRAM)



8 MECHANISM OF BREATHING

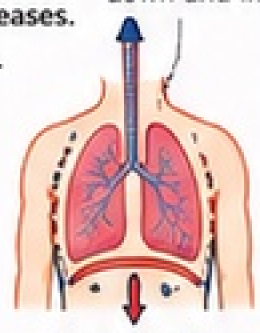
(A) BREATHING IN (INHALATION)

- Diaphragm contracts and moves downward.
- Rib cage moves upward and outward.
- Volume of thoracic cavity increases.
- Pressure inside lungs decreases.
- Air rushes in through nostrils into lungs.



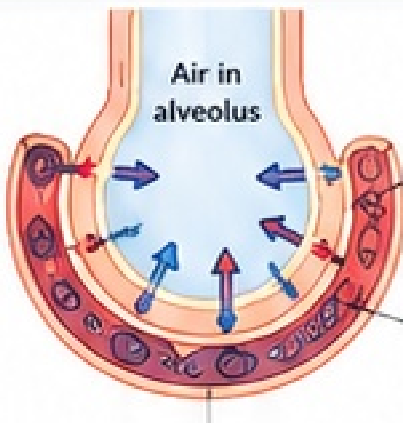
(B) BREATHING OUT (EXHALATION)

- Diaphragm relaxes and moves upward.
- Rib cage moves downward and inward.
- Volume of thoracic cavity decreases.
- Pressure inside lungs increases.
- Air is pushed out through nostrils.



9 GAS EXCHANGE IN ALVEOLI

- Alveoli are surrounded by a network of blood capillaries.
- Oxygen (O_2) from alveolar air diffuses into blood.
- Carbon dioxide (CO_2) from blood diffuses into alveoli.
- This happens by diffusion due to concentration gradient.
- Oxygen-rich blood is carried to body cells; CO_2-rich blood returns to lungs.



10 AEROBIC vs ANAEROBIC RESPIRATION

Feature	Aerobic Respiration	Anaerobic Respiration
Oxygen	Requires oxygen	Occurs in absence of oxygen
Site	Mitochondria	Cytoplasm
Breakdown of glucose	Complete	Incomplete
End products	$CO_2 + H_2O$	Ethanol + CO_2 (yeast) / Lactic acid (muscle cells)
Energy produced	Large amount (~36–38 ATP)	Small amount (~2 ATP)
Examples	Most body cells (liver, heart, brain, etc.)	Yeast (fermentation), Muscle cells during vigorous exercise
Odour	No smell	Ethanol gives characteristic smell (in yeast)

TRY THIS!

1. Explain why anaerobic respiration gives less energy.
2. What will happen if the diaphragm is paralysed?
(Hint: Think about breathing mechanism)

SECTION F – TRANSPORTATION IN HUMAN BEINGS: BLOOD, BLOOD VESSELS AND HEART

1. MEANING OF TRANSPORTATION

- **Transportation** is the process of carrying substances like oxygen, nutrients, hormones, gases, wastes, etc. from one part of the body to another.

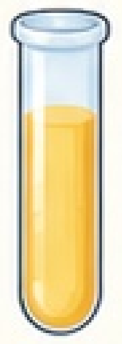
2. NEED FOR TRANSPORT IN MULTICELLULAR ORGANISMS

- In multicellular organisms, cells are far apart.
- Diffusion is not sufficient for all cells.
- A specialised transport system is required to deliver substances to every cell and remove wastes.

3. BLOOD AS FLUID CONNECTIVE TISSUE

- Blood is a special type of connective tissue.
- It is a **fluid** because of its liquid matrix called plasma.
- It has a **fluid** matrix and formed elements (RBCs, WBCs and platelets).

4. COMPONENTS OF BLOOD AND THEIR FUNCTIONS



(i) PLASMA (~55%)
- Straw-coloured liquid part.
- **Mostly water** (90–92%) with proteins, salts, glucose, amino acids, hormones, vitamins, urea, etc.
- **Function:** Transport of nutrients, hormones, gases; removal of wastes; maintains pH and osmotic balance; helps in clotting.

(ii) RED BLOOD CELLS (RBCs) / ERYTHROCYTES (~45%)
- Biconcave, no nucleus, contain haemoglobin.
- **Function:** Transport oxygen from lungs to tissues and some CO_2 from tissues to lungs.

(iii) WHITE BLOOD CELLS (WBCs) / LEUKOCYTES (<1%)
- Have nucleus, irregular shape, can move.
- **Function:** Body defence; destroy germs; immunity.

(iv) PLATELETS / THROMBOCYTES (<1%)
- Tiny cell fragments, no nucleus.
- **Function:** Help in blood clotting and prevent excessive blood loss.

5. BLOOD VESSELS AND THEIR FEATURES

- **Arteries** carry blood away from the heart.
- **Veins** carry blood towards the heart.
- **Capillaries** connect arterioles and venules.
- They are the **site** of exchange of materials.

COMPARISON OF BLOOD VESSELS

Feature	ARTERIES	VEINS	CAPILLARIES
Direction of blood flow	Away from heart	Towards heart	Connect arterioles to venules
Wall thickness	Thick, elastic, muscular	Thin, less elastic, less muscular	Very thin (one cell thick)
Lumen	Narrow	Wide	Very narrow
Valves	Absent	Present	Absent
Pressure	High	Low	Very low
Type of blood (general)	Usually oxygenated* (except pulmonary artery)	Usually deoxygenated* (except pulmonary veins)	May be oxygenated or deoxygenated

*Exceptions: Pulmonary artery – deoxygenated; Pulmonary veins – oxygenated.

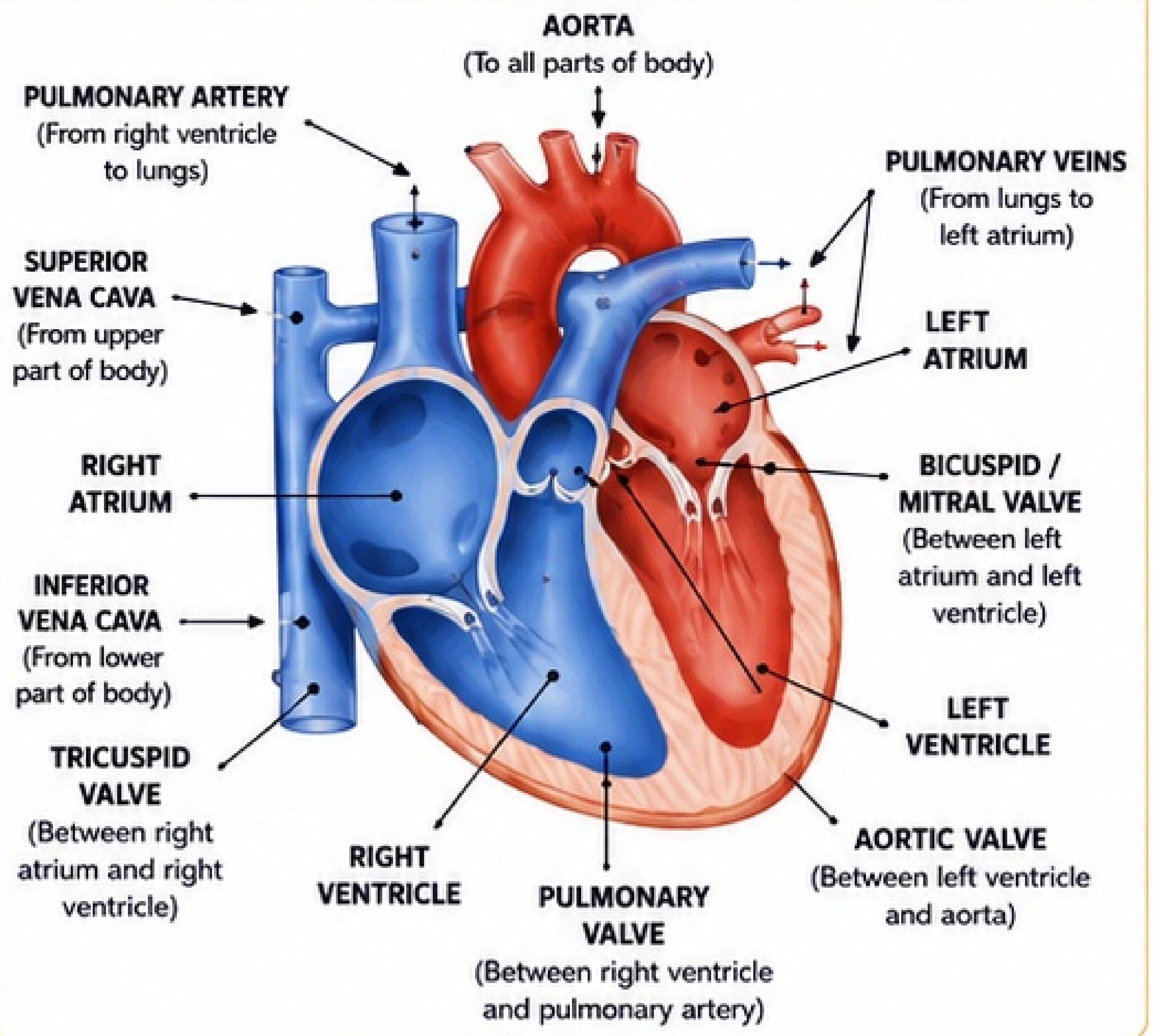
6. STRUCTURE AND FUNCTION OF HUMAN HEART

- Heart is a muscular, hollow organ, about the size of a fist.
- Located in the thoracic cavity, slightly left of the midline.
- It works as a double pump to circulate blood.

7. FOUR CHAMBERS OF THE HEART

- **Right atrium (RA)** – receives deoxygenated blood from body.
- **Right ventricle (RV)** – pumps deoxygenated blood to lungs.
- **Left atrium (LA)** – receives oxygenated blood from lungs.
- **Left ventricle (LV)** – pumps oxygenated blood to body.

8. HUMAN HEART – LABELLED DIAGRAM



9. DOUBLE CIRCULATION

- **Pulmonary Circulation (short circuit):**
  Right ventricle → Pulmonary artery → Lungs → Pulmonary veins → Left atrium
- **Systemic Circulation (long circuit):**
  Left ventricle → Aorta → Body parts → Venae cavae → Right atrium

10. IMPORTANCE OF SEPARATION OF OXYGENATED AND DEOXYGENATED BLOOD

- Prevents mixing of oxygenated and deoxygenated blood.
- Ensures efficient supply of oxygen to all body cells.
- Supports high energy level and proper functioning of body.

11. HEARTBEAT, PULSE AND BLOOD PRESSURE

- **Heartbeat:** One complete cycle of contraction and relaxation of the heart is called one heartbeat. Normal rate: 72–80 beats/min.
- **Pulse:** Rhythmic expansion and collapse of an artery due to blood pumped by the heart. Usually felt at wrist or neck. Normal pulse rate: 72–80 beats/min.
- **Blood Pressure:** Pressure exerted by blood on the walls of arteries. Normal blood pressure: ~120/80 mm Hg. (Systolic ~120 mm Hg, Diastolic ~80 mm Hg).

QUICK RECAP

- ✅ Blood = Plasma + RBCs + WBCs + Platelets
- ✅ Arteries → Away | Veins → Towards | Capillaries → Exchange
- ✅ Heart has 4 chambers and works as a double pump
- ✅ Double circulation = Pulmonary + Systemis

IF YOU SEE

- » Fatigue, breathlessness → May be low RBC / anaemia
- » Bleeding → Platelets help in clotting
- » Infection → WBCs fight germs
- » High BP → Risk for heart problems

IMPORTANT TERMS

- Plasma, Haemoglobin, Clotting
- Artery, Vein, Capillary
- Atrium, Ventricle, Valve
- Pulmonary Circulation
- Systemic Circulation

EXAM TIP

- Learn the heart diagram thoroughly.
- Do not mix pulmonary artery (deoxygenated) with pulmonary veins (oxygenated).
- Remember: **Arteries** → **Away**, **Veins** → **Towards**.

SECTION G - TRANSPORTATION IN PLANTS AND TRANSPIRATION

1. **Need for transport in plants:** Plants need to transport water, minerals from roots to leaves and food from leaves to all parts for growth, repair and other life activities.

2. **Xylem and phloem:** Xylem and phloem are the two main vascular tissues in plants that help in transport of substances.

3. **Xylem transports water and minerals:** Xylem is a complex tissue that conducts water and dissolved minerals from the roots to the stem and leaves. It also provides mechanical strength to the plant.

4. **Phloem transports food:** Phloem is a living tissue that transports prepared food (sugars) from leaves to other parts of the plant.

5. **Translocation:** The process of transport of food in phloem from leaves to other parts (upward, downward or sideways) is called translocation.

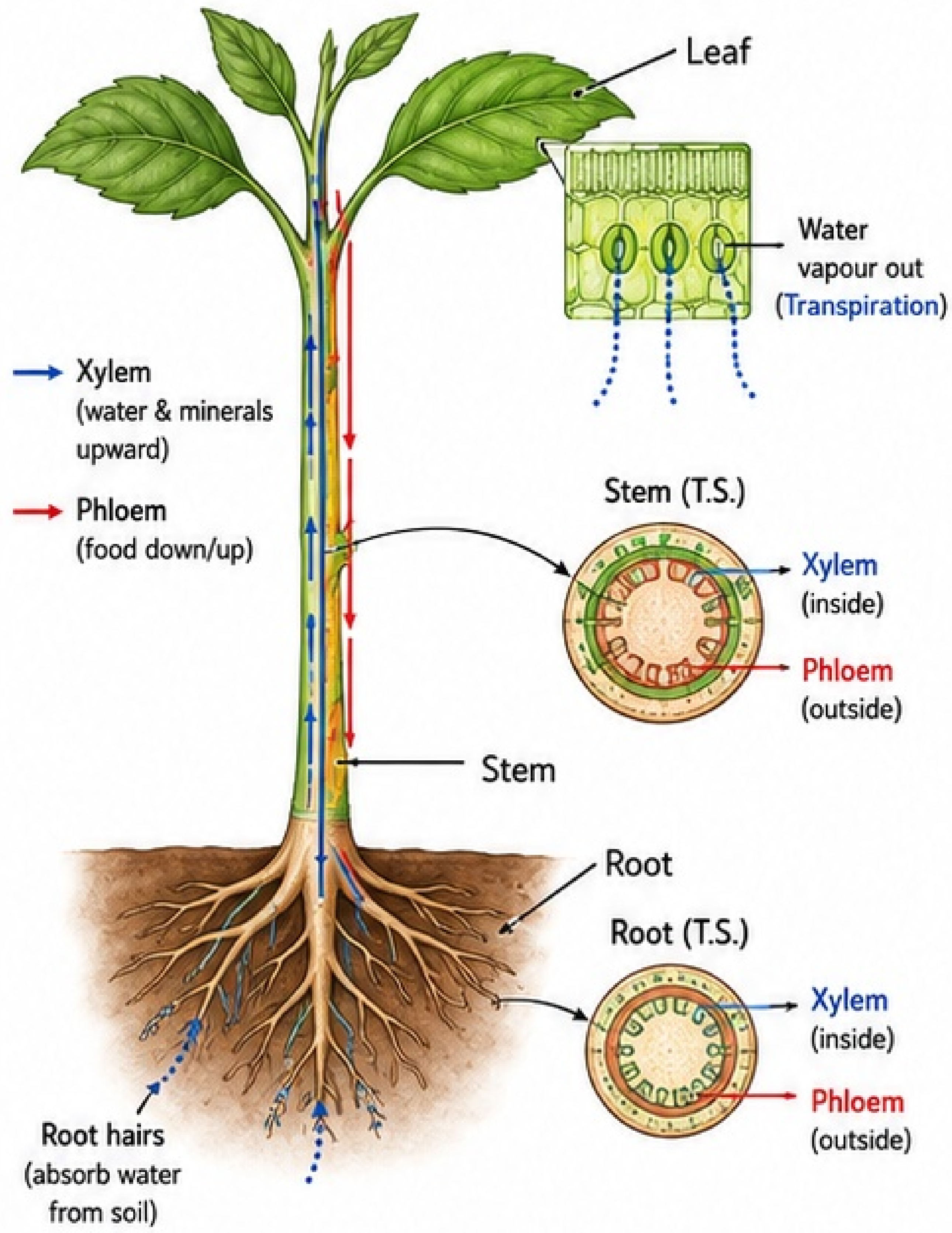
6. **Transpiration – meaning:** Transpiration is the loss of water in the form of water vapour from the aerial parts of plants, mainly through stomata in leaves.

   Significance: It helps in cooling the plant and creates a pull for upward movement of water and minerals.

7. **Root pressure and transpiration pull:**
   - **Root pressure:** Positive pressure developed in roots due to active absorption of water; pushes water upward.
   - **Transpiration pull:** Suction force created by transpiration from leaves; pulls water upward through xylem.

8. **Movement of water (pathway):** Water enters root hairs from soil → passes through cortex → endodermis → xylem of root → xylem of stem → xylem of leaf → mesophyll cells → evaporates through stomata.

9. PLANT TRANSPORT PATH (VASCULAR TISSUES)



Water and minerals move upward through xylem. Food is transported through phloem to all parts.

10. XYLEM vs PHLOEM

Feature	Xylem	Phloem
Nature	Mostly dead cells	Living cells
Main function	Transports water and minerals (upward)	Transports food (sugars) (to all parts)
Direction of transport	Unidirectional (upward)	Bidirectional (up and down)
Cells involved	Tracheids, vessels, fibres, parenchyma	Sieve tubes, companion cells, parenchyma, fibres
Cell wall	Thick, lignified	Thin, non-lignified
Support role	Provides mechanical strength	No significant role in support

11. ROLE OF STOMATA IN TRANSPIRATION

- Stomata are tiny pores present on the surface of leaves.
- They regulate the loss of water vapour during transpiration.
- Guard cells open and close stomata.
- More stomata → more transpiration.
- During night or in dry conditions, stomata close to reduce water loss.

12. IMPORTANCE OF TRANSPIRATION

- Causes cooling of the plant, preventing overheating.
- Creates transpiration pull that helps in upward movement of water and minerals from roots to leaves.
- Helps in maintaining water balance in plants.
- Necessary for the transport of dissolved substances in the plant.

CLASS 10

Ch 5: Life Processes

SECTION H – EXCRETION IN HUMAN BEINGS

1. Meaning of Excretion

- Excretion is the biological process by which the body removes **metabolic wastes** produced during life processes.

2. Excretion vs Egestion

- **Excretion** removes waste produced inside the cells during metabolism (e.g., urea, CO_2, water, salts).
- **Egestion** removes undigested and unabsorbed food from the alimentary canal.

3. Importance of Removing Metabolic Wastes

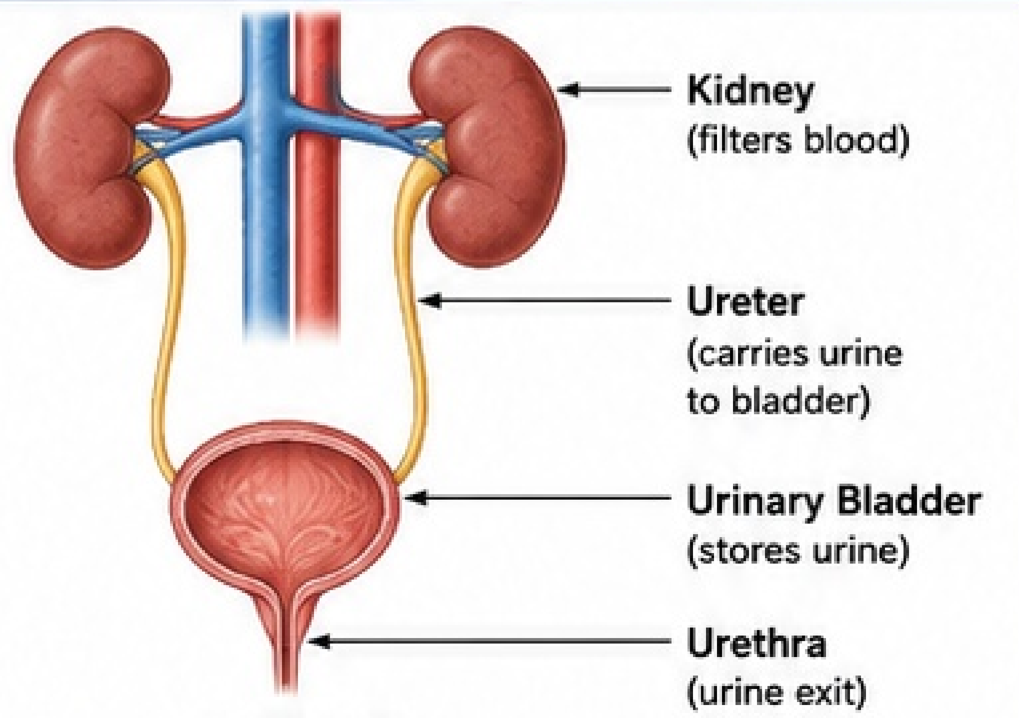
- Prevents accumulation of toxic substances.
- Maintains water, salt and pH balance (homeostasis).
- Ensures proper functioning of cells, tissues and organs.
- Helps in regulation of osmoregulation and blood volume.

4. Human Excretory System and Its Parts

The human **excretory system** (urinary system) consists of:

- **Kidneys** (pair): Filter blood and form urine.
- **Ureters** (pair): Carry urine from kidneys to urinary bladder.
- **Urinary Bladder**: Temporarily stores urine.
- **Urethra**: Carries urine from urinary bladder to outside.

5. Diagram: Human Urinary System (Labeled)

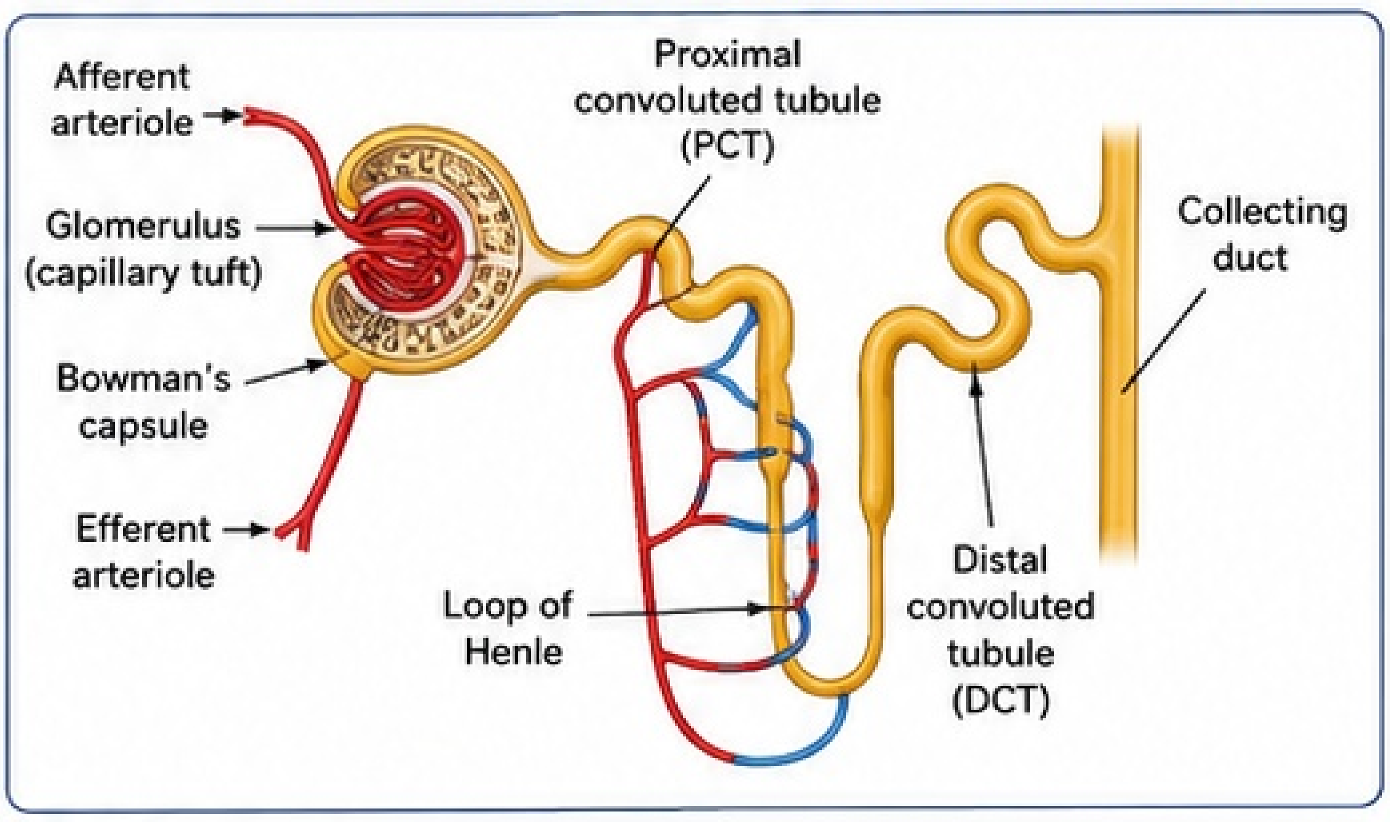


(Front view)

9. Dialysis / Artificial Kidney

- **Dialysis** is a process that cleans the blood when kidneys fail to do so.
- A dialysis machine (artificial kidney) filters wastes, excess salts and extra water from blood through a semipermeable membrane.
- Needed in cases of kidney failure, chronic kidney disease, or severe damage to kidneys.
- **Types:** Haemodialysis (most common), Peritoneal dialysis.

6. Structure of Nephron (Functional Unit of Kidney)



7. Steps of Urine Formation

(1) Ultrafiltration (Glomerular filtration)

- High blood pressure in glomerulus forces water and small **solutes (urea, salts**, glucose, amino acids, etc.) through the filtration membrane into Bowman's capsule.
  Cells and plasma proteins do not pass.

(2) Selective Reabsorption

- Useful substances like glucose, amino acids, water and required ions are reabsorbed from the filtrate back into the blood mainly in PCT (and loop of Henle, DCT).

(3) Tubular Secretion

- Certain wastes and excess ions (H^+, K^+, NH_4^+, drugs, toxins) are secreted from blood into the filtrate in PCT and DCT to maintain proper composition.

8. Role of Kidneys

- Remove nitrogenous wastes (mainly urea) from blood.
- Regulate water and electrolyte balance → osmoregulation.
- Maintain acid–base balance (pH) of blood.
- Help in control of blood volume and blood pressure (via **renin** hormone).
- Activate vitamin D and produce erythropoietin (RBC formation).

10. Excretion vs Egestion (Comparison Table)

Feature	Excretion	Egestion
Meaning	Removal of metabolic wastes produced in the body.	Removal of undigested and unabsorbed food.
Nature of Material	Metabolic, toxic, soluble wastes (urea, CO_2, water, salts).	Non-digestible, solid, unabsorbed food.
Site/Organ	Kidneys, lungs, skin, liver (urinary system mainly).	Large intestine (alimentary canal).
Mode	By specific excretory organs (e.g., kidney).	Through anus after defecation.
Purpose	Maintains internal environment (homeostasis).	Eliminates food that was never absorbed.
Example	Urea in urine, CO_2 in breath, sweat, excess salts.	Fibre, undigested food residues.

11. Pathway: From Blood to Urine Exit (Sequence Chart)

Blood in Renal Artery (from aorta) → Afferent Arteriole → Glomerulus (Filtration) → Bowman's Capsule → Renal Tubule (PCT → Loop of Henle → DCT) → Collecting Duct → Ureter → Urinary Bladder → Urethra → Outside (Urine Exit)

EXAM TIP: Draw neat, labeled diagrams. Learn differences (Excretion vs Egestion). Write steps of urine formation in order. Focus on functions of kidney.

CENTUM FOCUS: Understand + **Recall**
Diagrams + Tables + Flow = FULL MARKS

Ch 5: Life Processes

STUDY SMART SCORE CENTUM!

SECTION I - EXCRETION IN PLANTS, IMPORTANT COMPARISONS AND MUST-KNOW DIAGRAMS

1 EXCRETION IN PLANTS

- Plants do not have a special excretory system. Wastes are removed or stored in different parts or released in various forms.

2 GASEOUS WASTE RELEASE THROUGH STOMATA AND LENTICELS

- O_2 produced in photosynthesis and CO_2 from respiration are exchanged through stomata (in leaves) and lenticels (in stems).

3 STORAGE OF WASTES IN LEAVES, BARK, VACUOLES

- Some wastes are stored in old leaves, bark, and in vacuoles of cells and then removed when these parts fall off.

4 EXCRETION AS RESINS, GUMS, LATEX

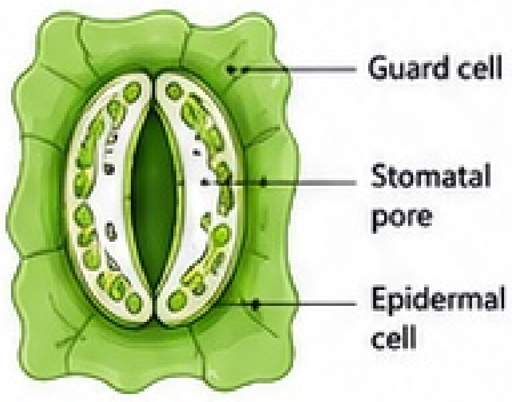
- Many plants remove wastes in the form of resins (e.g., pine), gums (e.g., acacia) and latex (e.g., rubber, calotropis).

5 EXCESS WATER REMOVAL BY TRANSPIRATION

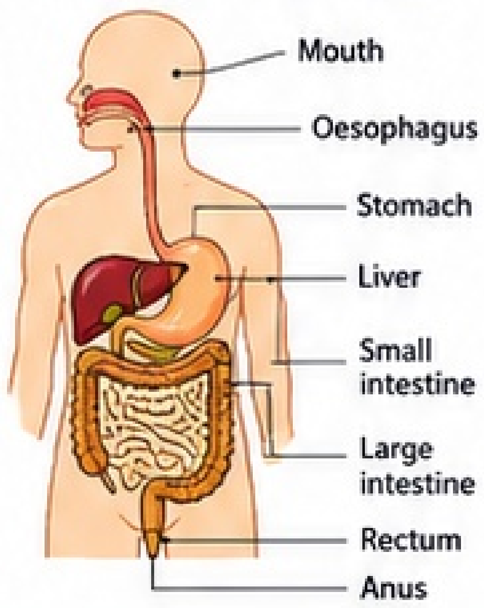
- Excess water absorbed by roots is removed as water vapour through stomata in a process called transpiration.

MUST-KNOW DIAGRAMS (QUICK RECOGNITION)

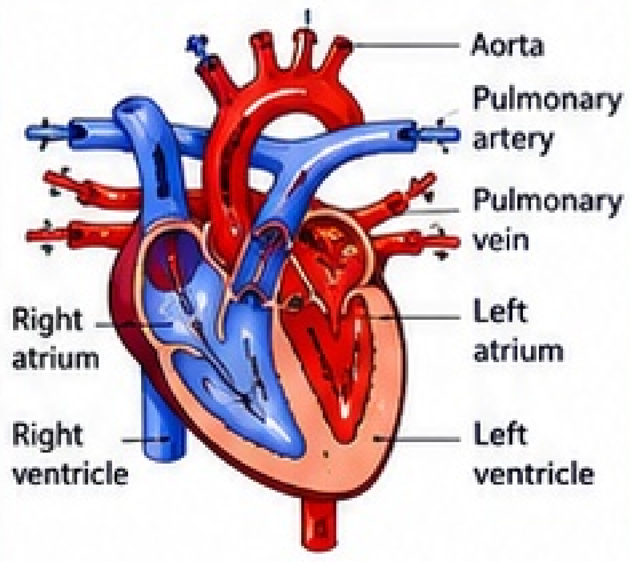
1. STOMATA



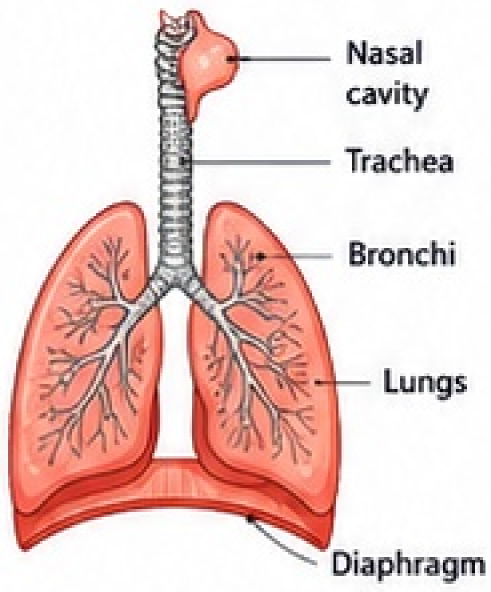
2. DIGESTIVE SYSTEM



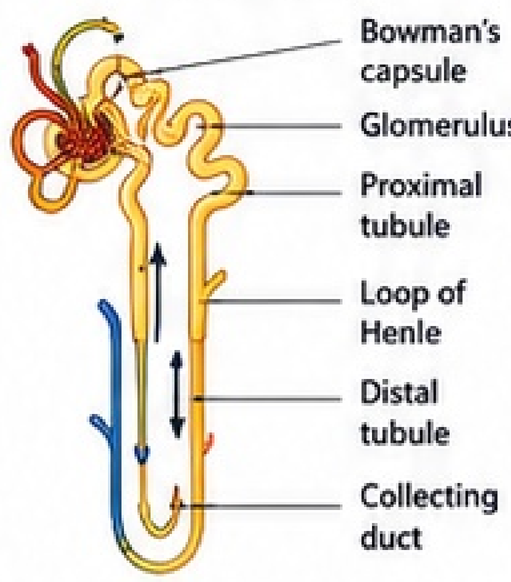
3. HEART / DOUBLE CIRCULATION



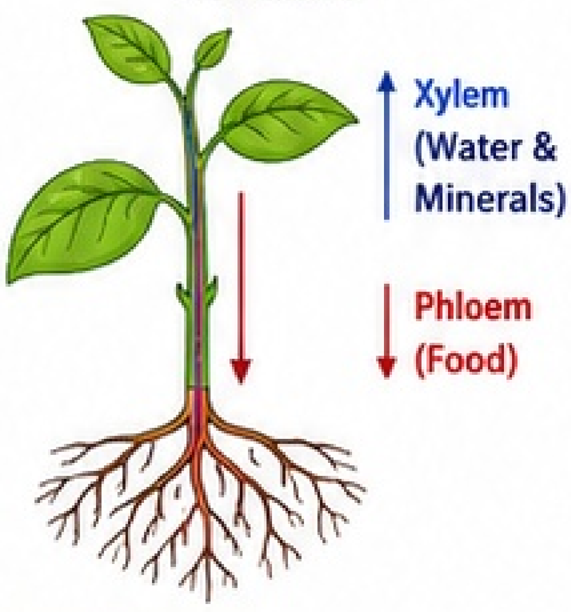
4. RESPIRATORY SYSTEM



5. NEPHRON (STRUCTURAL UNIT OF KIDNEY)



6. TRANSPORT PATHWAY IN PLANTS



IMPORTANT COMPARISONS (MUST-KNOW TABLES)

1 BREATHING vs RESPIRATION

	Breathing	Respiration
Nature	Physical process	Biochemical process
Purpose	Inhalation and exhalation of air	Release of energy from food
Site	Lungs	Cells (via mitochondria)
Outcome	Air exchange	Energy (ATP) + CO_2 + H_2O

2 AEROBIC vs ANAEROBIC RESPIRATION

	Aerobic (With O_2)	Anaerobic (Without O_2)
Oxygen	Required	Not required
Site	Mitochondria	Cytoplasm
End products	CO_2 + H_2O	Lactic acid or Ethanol + CO_2
Energy	High (36–38 ATP)	Low (2 ATP)

3 AUTOTROPHIC vs HETEROTROPHIC NUTRITION

	Autotrophic	Heterotrophic
Definition	Make own food	Depends on others for food
Source	CO_2, H_2O, sunlight	Plants/Animals
Example	Green plants, some bacteria	Human beings, animals
Mode	Photosynthesis	Ingestion / Absorption

4 XYLEM vs PHLOEM

	Xylem	Phloem
Function	Transports water & minerals	Transports food
Direction	Upward only	Upward & downward
Cells	Dead (except xylem parenchyma)	Living
Elements	Tracheids, Vessels, Xylem parenchyma	Sieve tubes, Companion cells, Phloem parenchyma

5 ARTERY vs VEIN

	Artery	Vein
Direction	Away from heart	Towards heart
Blood (mostly)	Oxygenated	Deoxygenated
Wall	Thick, elastic	Thin, less elastic
Pressure	High	Low
Valve	Absent	Present

6 EXCRETION vs EGESTION

	Excretion	Egestion
Definition	Removal of metabolic wastes	Removal of undigested food
Waste type	Urea, uric acid, CO_2, excess water, etc.	Fibre, undigested residues
Site in human	Kidneys, lungs, skin, liver	Large intestine (via anus)
Purpose	Maintains internal balance	Cleans digestive tract

SHORT ANSWER – ANSWER FRAMES (BOARD STYLE REVISION)

1 Define photosynthesis.

- Photosynthesis is the process by which ________ ____________________
- It occurs in ____________ in the presence of __________.

2 State the function of xylem.

- Xylem is the vascular tissue that ____________________ ____________________
- It transports ____________ ____________________.

3 Define nephron.

- Nephron is the structural and functional unit of the kidney.
- It consists of ____________ ____________________ ____________________.

4 Explain double circulation.

- Double circulation means blood passes through the heart ____________ in one complete cycle.
- It includes pulmonary and systemic circulation.

5 Name the wastes removed by kidneys.

- Kidneys remove ____________, ____________, ____________, and excess ____________ from the blood.

Ch 5: Life Processes

SECTION J – MASTER REVISION SHEET, CONCEPT MAP, NCERT TRAPS AND CENTUM FOCUS

1. KEY DEFINITIONS

	Term	Definition
1.	Life process	The basic activities performed by living organisms to maintain life.
2.	Nutrition	The process by which organisms take in food and utilise it for energy, growth and repair.
3.	Autotrophic nutrition	Mode of nutrition in which organisms (e.g., green plants) make their own food from simple inorganic substances.
4.	Heterotrophic nutrition	Mode of nutrition in which organisms depend on other organisms for food.
5.	Respiration	The process of breakdown of food inside the cells to release energy.
6.	Transportation	The process of movement of substances (food, gases, water, wastes, hormones) within the body.
7.	Excretion	The process of removal of metabolic wastes produced in the body.
8.	Photosynthesis	The process by which green plants make food using sunlight, chlorophyll, CO_2 and water and release O_2.
9.	Transpiration	The loss of water in the form of water vapour from the aerial parts of plants.
10.	Nephron	The structural and functional unit of the kidney responsible for filtration of blood and formation of urine.
11.	Double circulation	The condition in which blood passes through the heart twice during one complete circulation of the body.

2. MASTER CHAPTER SUMMARY

- Life processes are essential for survival.
- Nutrition:
  - Autotrophic (plants) – prepare food by photosynthesis.
  - Heterotrophic (animals, fungi, most bacteria) – depend on other organisms.
- Respiration:
  - Aerobic respiration releases energy completely with O_2.
  - Anaerobic respiration occurs in the absence of O_2 (yeast, some bacteria, muscles during exercise).
- Transportation:
  - In plants: Xylem (water & minerals upward), Phloem (food to all parts; bidirectional).
  - In animals: Blood, Heart and Blood vessels constitute the transport system.
- Excretion:
  - Plants: CO_2, O_2, excess water, gums, resins, etc.
  - Humans: Kidneys form urine to remove nitrogenous wastes.
- Kidneys contain millions of nephrons which filter blood and form urine.
- In humans, **double circulation** keeps oxygenated and deoxygenated blood separate, ensuring efficient supply of O_2.

3. CONCEPT MAP

NUTRITION (Obtaining & utilising food)
- Provides energy
- Provides materials for growth & repair

RESPIRATION (Breakdown of food in cells)
- Releases energy
- Occurs in cells
- Aerobic / Anaerobic

LIFE PROCESSES

TRANSPORTATION (Movement of substances)
- In plants: Xylem & Phloem
- In animals: Blood, Heart, Blood vessels
- Delivers what is needed, removes what is not needed

EXCRETION (Removal of waste products)
- Removes metabolic wastes
- Maintains internal balance
- In humans: Kidneys, Nephrons, Urine

These processes are interdependent and maintain life and homeostasis in organisms.

4. NCERT TRAPS / COMMON MISCONCEPTIONS

	Trap / Misconception	The Truth
1	Breathing is the same as respiration.	Breathing is a physical process of taking in and giving out air. Respiration is a biochemical process of breaking down food in cells to release energy.
2	Xylem is a living tissue.	Xylem is mostly made of dead cells. Only xylem parenchyma is living.
3	Phloem transports food only downward.	Phloem transport can be upward or downward depending on the source and sink (bidirectional).
4	Blood in arteries is always oxygenated.	Generally yes, but pulmonary artery carries deoxygenated blood from heart to lungs.
5	Kidneys filter food and make urine.	Kidneys filter blood, not food. After absorption of nutrients in the intestine, the blood is filtered in kidneys to remove wastes.
6	Excretion and egestion are the same.	Excretion removes metabolic wastes from cells (e.g., urea, CO_2). Egestion is the removal of undigested food from the body.

5. 10 MUST REMEMBER POINTS

1. Nutrition provides energy and materials.
2. Photosynthesis needs light, chlorophyll, CO_2 and water.
3. Respiration releases energy from food.
4. Aerobic respiration is more efficient than anaerobic.
5. Xylem transports water and minerals upward.
6. Phloem transports food to all parts (bidirectional).
7. Double circulation keeps oxygenated and deoxygenated blood separate.
8. Kidneys contain millions of nephrons.
9. Excretion maintains internal balance and prevents accumulation of wastes.
10. Life processes work together to sustain life and maintain homeostasis.

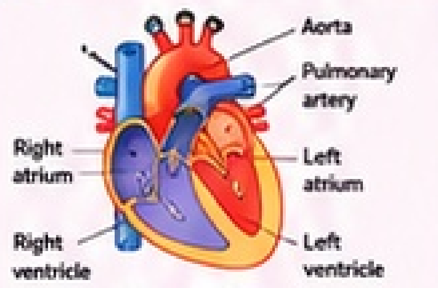
6. CENTUM CHECKLIST

✓	Task
☐	I can define all key terms correctly.
☐	I can explain autotrophic and heterotrophic nutrition with examples.
☐	I can explain respiration (aerobic & anaerobic) with examples.
☐	I can draw and label the human heart and nephron neatly.
☐	I can explain transportation in plants and animals clearly.
☐	I can differentiate between excretion and egestion.
☐	I can answer NCERT intext and exercise questions without mistakes.
☐	I can write diagram-based answers correctly.
☐	I can handle case-based and application based questions.
☐	I can write short notes in perfect points.

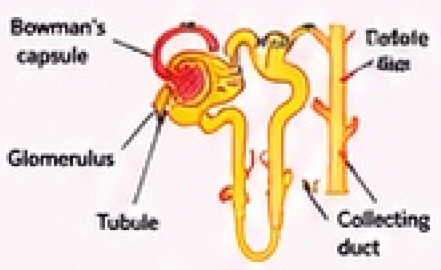
7. BEFORE-EXAM QUICK GLANCE

CONCEPTS
- Nutrition (Auto / Hetero)
- Photosynthesis
- Respiration (Aerobic/ Anaerobic)
- Transportation in plants & animals
- Excretion in plants & humans
- Nephron and urine formation
- Double circulation

DIAGRAMS TO MASTER



Human Heart



Nephron

IMPORTANT TABLES

Functions of:
- Xylem vs Phloem
- Aerobic vs Anaerobic Respiration
- Excretion in Plants vs Humans

Blood Vessels:

Blood Vessel	Carries	Wall	Valves
Arteries	Away from heart	Thick, elastic	No
Veins	Towards heart	Thin, less elastic	Yes
Capillaries	Between arteries & veins	Very thin (1 cell thick)	No

KEY TERMS
- Nutrition
- Photosynthesis
- Respiration
- Transportation
- Excretion
- Xylem, Phloem
- Nephron
- Urea
- Double circulation
- Homeostasis

FORMULAS / FACTS
- Photosynthesis: $6CO_2 + 6H_2O$ (light, chlorophyll) $\rightarrow C_6H_{12}O_6 + 6O_2$
- Cellular respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP)
- Normal human pulse rate: 72–80 beats/min
- Adult kidneys: ~1–1.2 million nephrons in each

LAST MINUTE TIPS
- Read questions carefully.
- Use keywords from question in answers.
- Draw neat, labeled diagrams.
- Write in points for short answers.
- Manage time: easy questions first.
- Revise NCERT line by line – high score secret!